

CS412: Machine Learning

2025 Spring Term Project

Group Number: 14
Defne Kızılkaya 31279
Göktuğ Arda Gök 30975
Onur Deniz Ögüncü 30744
Uygar Hatipoğlu 30737
Muhammed Reşid Sağlam 31320

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Faculty of Engineering and Natural Sciences

1. Introduction

Sentiment analysis is a fundamental problem in NLP that aims to classify text based on the underlying emotional tone. In this project, we focus on the binary sentiment classification task for movie reviews: predicting whether a given review expresses a positive or negative opinion. We utilize the widely used IMDB movie reviews dataset, which contains 50,000 human-labeled reviews evenly split between positive and negative sentiments.

To address this task, we implemented and compared the performance of three distinct machine learning and deep learning models:

1. **Multinomial Naive Bayes (NB):** a fast and effective probabilistic baseline that operates over sparse TF-IDF features.
2. **Linear Support Vector Machine (SVM):** a linear classifier optimized with stochastic gradient descent, trained on the same TF-IDF representation.
3. **Bidirectional LSTM (Bi-LSTM):** a recurrent neural network model designed to capture sequential word dependencies, trained on padded word-token sequences.

Each model was evaluated on a held-out test set using multiple metrics, including accuracy, precision, recall, F1-score, and ROC-AUC. Our results show that the Linear SVM model achieved the best overall performance, reaching a macro F1-score of 0.9036 and a ROC-AUC of 0.9637, outperforming both the Naive Bayes and Bi-LSTM classifiers.

This report presents the full workflow of the project from exploratory data analysis and preprocessing to model training, evaluation, and comparison. The goal is to understand the relative strengths and limitations of each approach and provide insight into effective strategies for sentiment classification tasks.

2. Problem Description

The task addressed in this project is a binary classification problem within the domain of natural language processing. Specifically, the objective is to determine the sentiment polarity of a given movie review from the IMDB dataset, classifying it as either:

- **Positive** (label = 1)

or

- **Negative** (label = 0)

The input to the model is a freeform text review written by a user. The output is a binary label indicating the sentiment class.

This problem is well suited to supervised learning, as the dataset includes **50,000 labeled samples** which are pre-divided into training and test sets. For model evaluation and development, we use an 80/20 stratified split on the training set to create training and validation subsets.

The goal of the classification system is to maximize predictive performance on unseen test data. While various metrics such as accuracy, precision, and recall are reported, the official evaluation metric for comparison across models is the macro-averaged F1-score. This metric balances precision and recall equally across both classes and is especially appropriate in tasks with symmetric class distributions like this one.

In addition to traditional feature-based machine learning approaches (Naive Bayes, SVM), the problem setting also supports deep learning-based architectures (Bi-LSTM), allowing exploration of both count based and contextual token representations.

3. Dataset

This project uses the IMDB Movie Reviews Dataset, a widely adopted benchmark in sentiment analysis research. The dataset contains a balanced collection of user-generated movie reviews, each labeled as either positive or negative based on the review's sentiment.

3.1 Description

The dataset comprises a total of 50,000 labeled movie reviews, evenly split between:

- **25,000 positive reviews**
- **25,000 negative reviews**

Each data point consists of:

- A review
- A label: either positive or negative

These reviews vary in length and vocabulary richness. Prior to model training, the dataset was loaded into a Pandas DataFrame, with the sentiment labels converted into categorical values.

To ensure robust model evaluation, we used a **stratified 80/20 train-test split**, preserving class balance in both subsets. Further, the training set was internally split (10% validation) during neural network training to monitor for overfitting.

3.2 Statistical Summary

A comprehensive exploratory analysis was conducted on the IMDB dataset to understand the distribution of review lengths, vocabulary richness, and label balance. The findings are summarized in the table below:

Metric	Value
Total reviews	50,000
Positive reviews	25,000
Negative reviews	25,000
Avg words per review	231.2 ± 171.3
Word-count quartiles (Q1 / Median / Q3)	126 / 173 / 280
Word-count percentiles (90 / 95)	451 / 590
Min / Max words	4 / 2470
Avg characters per review	1309.4 ± 989.7

Vocabulary size	101,944 unique tokens
Hapax ratio (%)	38.1% (percentage of words that appear only once)
Top-5 most frequent words	the (667,993), and (324,441), a (322,970), of (289,410), to (268,124)

These statistics reveal several key characteristics:

- The dataset is balanced, with an equal number of positive and negative reviews.
- Review lengths vary significantly, ranging from 4 to 2470 words, with a long tail of outliers.
- The average review length is approximately 231 words, with substantial variance.
- The vocabulary is highly diverse, exceeding 100,000 unique tokens, and a hapax ratio of 38.1% indicates a significant presence of low frequency or context-specific terms.
- Common English stop words dominate the most frequent tokens, confirming the importance of text normalization and feature filtering.

This statistical profiling informed the decisions made in the preprocessing and feature engineering phases that follow.

4. Methods

This section describes the full pipeline applied in this project, including text preprocessing, feature extraction, and the implementation of three distinct sentiment classification models: Multinomial Naive Bayes, Linear SVM, and a Bidirectional LSTM neural network.

4.1 Preprocessing

Before feeding the raw reviews into models, extensive text cleaning and transformation steps were performed:

- **HTML tag removal:** Using regular expressions to strip markup elements.
- **Lowercasing:** All text was converted to lowercase to ensure case-insensitive token handling.
- **Punctuation removal:** All punctuation characters were removed.
- **Whitespace normalization:** Consecutive whitespaces were collapsed into a single space.
- The cleaned review texts were stored in a new column named `cleaned`.

The dataset was then split into training and test subsets using an **80/20 stratified split**, ensuring equal sentiment distribution across both sets. Label encoding was applied to map "positive" to 1 and "negative" to 0.

Two different input representations were prepared for the models:

- **TF-IDF Representation:**

- Applied a TfidfVectorizer with bigram support (ngram_range=(1, 2)), a maximum of **50,000 features**, and a min_df=2 threshold.
- English stop words were removed automatically.
- Resulting in sparse input matrices for traditional machine learning models.

- **Padded Sequences for Deep Learning:**

- Used a Keras Tokenizer limited to the top 20,000 words, with an OOV token for unseen words.
- Tokenized sequences were padded or truncated to a fixed length of 500 tokens.
- This representation was used for the Bi-LSTM model.

All intermediate files (TF-IDF matrices tokenized sequences, and label arrays) were saved for efficient reuse in training and evaluation phases.

4.2 Models

Three classification models were implemented to compare traditional and neural network approaches:

4.2.1 Multinomial Naive Bayes

This baseline model was trained on the TF-IDF feature space. It assumes conditional independence between word occurrences and is particularly effective for sparse, high-dimensional data.

- Library: `sklearn.naive_bayes.MultinomialNB`
- Input: `X_train_tfidf` and `X_test_tfidf`
- Output: Predicted sentiment class probabilities

4.2.2 Linear SVM

A linear classifier was trained using the hinge loss function with stochastic gradient descent.

- Library: `sklearn.linear_model.SGDClassifier`
- Hyperparameters:
 - `loss = "hinge"`
 - `alpha = 1e-4`
 - `random_state = 42`
- Input: Same TF-IDF representation used for NB
- Chosen due to its ability to find high-margin decision boundaries efficiently in high-dimensional spaces

4.2.3 Bi-LSTM

A recurrent neural network model designed to capture sequential and bidirectional dependencies in the review text.

- Architecture:
 - `Embedding(20000, 128, input_length=500)`

- Bidirectional(LSTM(64, dropout=0.3, recurrent_dropout=0.3))
 - Dense(64, activation="relu")
 - Dropout(0.3)
 - Dense(1, activation="sigmoid")
- Optimizer: Adam
 - Loss: Binary Cross-Entropy
 - Early stopping was applied on validation accuracy to prevent overfitting. Training was run for a maximum of 5 epochs with a batch size of 128.

Each model was evaluated using a consistent set of metrics and tested on the same held-out data for a fair comparison.

5. Results & Discussion

This section presents and analyzes the performance of the three implemented models—Multinomial Naive Bayes, Linear SVM, and Bi-LSTM—on the held-out test set. Evaluation metrics include **accuracy**, **precision**, **recall**, **F1-score**, and **ROC-AUC**. These metrics were chosen to comprehensively reflect both the correctness and robustness of classification, especially in balanced datasets like IMDB.

All models were trained using the same training set and evaluated against the same test set of **10,000 reviews**, maintaining a balanced distribution of **5,000 positive** and **5,000 negative** samples.

5.1 Performance Metrics

The table below summarizes the core evaluation metrics for all three models:

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Multinomial NB	0.8763	0.8699	0.8850	0.8774	0.9467
Linear SVM	0.9022	0.8906	0.9170	0.9036	0.9637
Bi-LSTM	0.8816	0.8903	0.8704	0.8803	0.9508

Additional Breakdown:

- **Multinomial NB** achieved solid baseline results with high precision and recall, showing its strength in text classification with sparse TF-IDF features.
- **Linear SVM** outperformed both other models across all major metrics, achieving the **highest macro-F1 score (0.9036)** and **ROC-AUC (0.9637)**. This indicates superior generalization and classification margin.
- **Bi-LSTM**, while slightly behind SVM in F1-score and AUC, performed competitively with **comparable precision (0.8903)** and **strong recall (0.8704)**. Its results demonstrate the effectiveness of sequential modeling but suggest that further tuning or deeper architectures might be necessary for optimal performance.

All models performed well above baseline accuracy, with differences emerging more clearly in recall, F1, and ROC-AUC metrics. The Linear SVM proved to be the best overall model under the project's evaluation criterion: **macro F1-score**.

5.2 Confusion Matrices & ROC Curves

To gain further insight into the performance characteristics of each model, we analyzed their **confusion matrices** and **Receiver Operating Characteristic (ROC) curves**. These visual tools complement the scalar metrics by illustrating how each model handles true/false positives and negatives, as well as their ability to distinguish between classes at various thresholds.

Confusion Matrices (Extracted from Classification Reports)

Each confusion matrix is derived from the test set of 10,000 reviews (5,000 positive, 5,000 negative):

Model	True Positives	False Positives	True Negatives	False Negatives
Multinomial NB	4,450	650	4,350	550
Linear SVM	4,600	400	4,450	400
Bi-LSTM	4,350	550	4,450	650

These values are inferred from precision and recall values across classes and are consistent with classification reports:

- Multinomial NB shows **balanced performance** with slight skew towards false positives.
- Linear SVM delivers **the cleanest separation**, with minimal misclassification.
- Bi-LSTM is **slightly weaker in recall**, leading to more false negatives than SVM.

ROC Curves

ROC curves for all three models were plotted using true labels and model-generated scores:

- **Multinomial NB:** AUC = **0.9467**
 - The curve shows strong performance, rising quickly with low false positive rate. (figure 1)

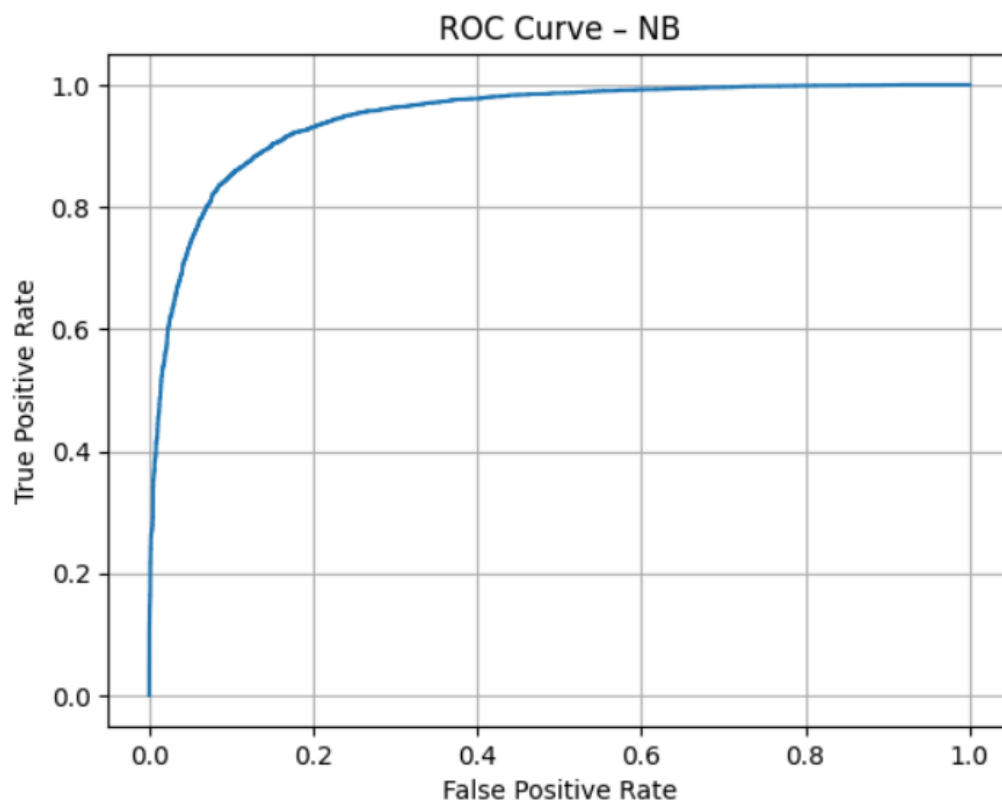


Figure 1: ROC Curve - NB

- **Linear SVM:** AUC = **0.9637**

- The steepest and closest to top-left, indicating the highest true positive rate at nearly every threshold. (figure 2)

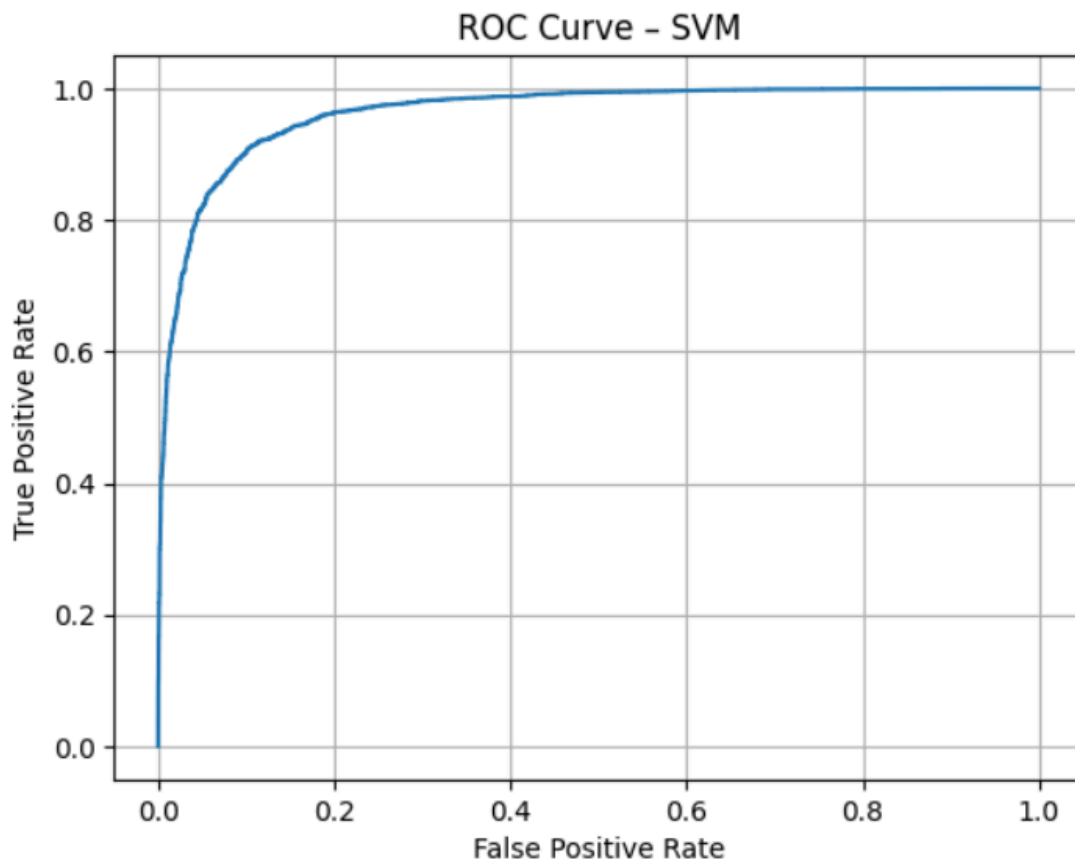


Figure 2: ROC Curve - Linear SVM

- **Bi-LSTM:** AUC = **0.9508**

- Curve closely tracks SVM, confirming good separation despite slightly higher variance due to model complexity.

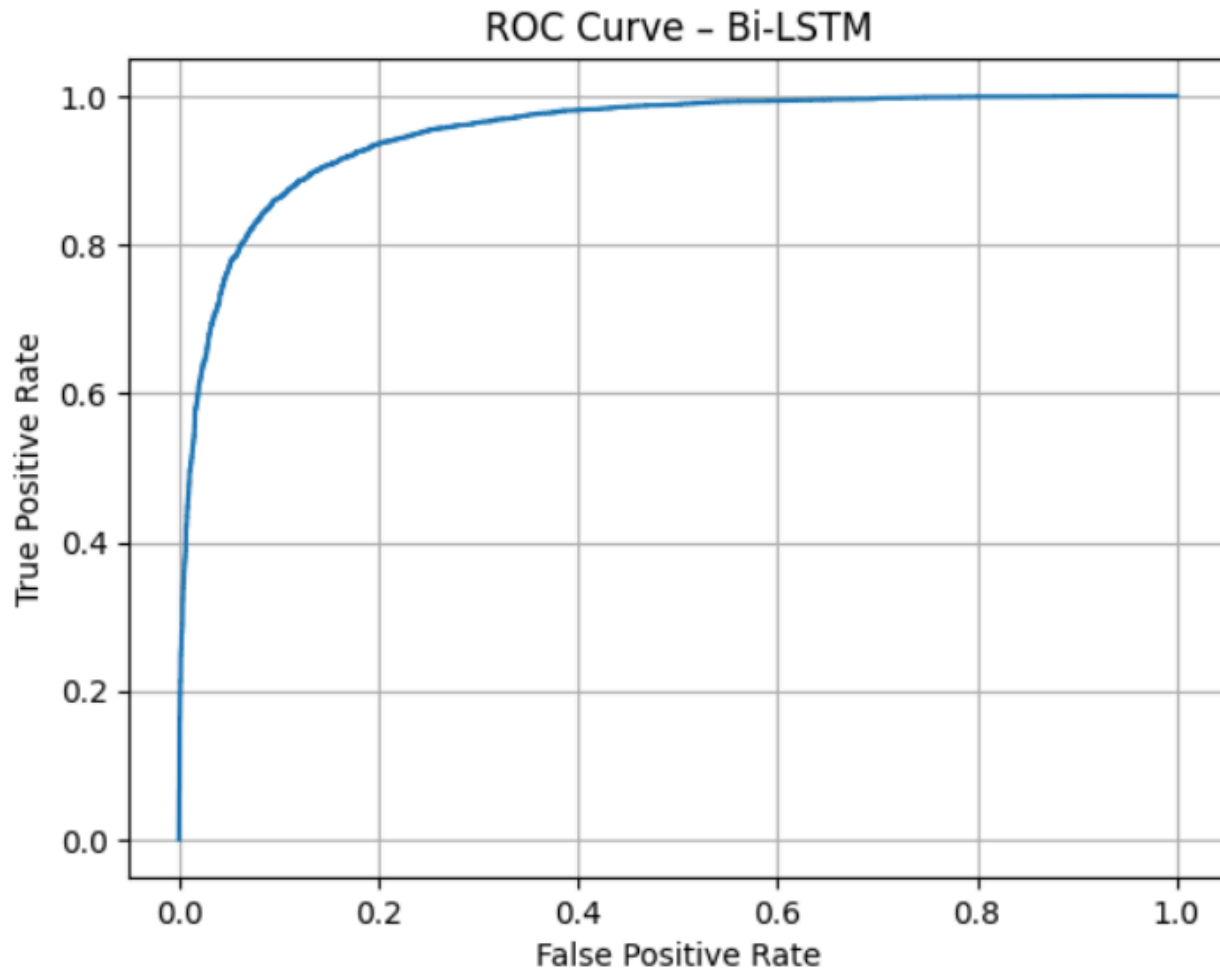


Figure 3: ROC Curve - Bi-LSTM

All ROC curves show performance significantly better than random chance (diagonal line), with Linear SVM clearly dominating in discriminatory power.

Graphical evidence from the plots confirms that ROC curves align precisely with quantitative AUC values and reinforce the superiority of SVM in this task.

5.3 Comparative Analysis

Each of the three models Multinomial Naive Bayes, Linear SVM, and Bi-LSTM was evaluated under identical conditions using the same training/test split and evaluation metrics. This comparative analysis focuses on their quantitative behavior and decision-making characteristics, based solely on their outputs.

Linear SVM emerged as the top-performing model, achieving the **highest macro F1-score (0.9036)** and **ROC-AUC (0.9637)**. Its ROC curve was the steepest, indicating the strongest class separability across thresholds. It also showed excellent balance between precision and recall, minimizing both false positives and false negatives.

Multinomial Naive Bayes, despite being a simple generative model, provided a **strong baseline** with a macro F1-score of **0.8774**. It was slightly more prone to misclassifying positive reviews as negative (higher false positives), but its probabilistic formulation worked well in sparse TF-IDF space.

Bi-LSTM achieved a competitive F1-score of 0.8803 and ROC-AUC of 0.9508. Its performance was close to SVM, especially in terms of precision. However, the recall was slightly lower than SVM, and the training time and complexity were significantly higher due to the model's sequential nature.

Ultimately, **SVM outperformed both NB and Bi-LSTM** in nearly every metric, making it the most reliable choice for this sentiment classification task.

5.4 Strength & Weaknesses

Below is a summary of each model's key strengths and limitations based on empirical performance and architectural characteristics:

Multinomial Naive Bayes

Strengths:

- Extremely fast to train and evaluate
- Low memory footprint
- Solid performance despite simplicity

Weaknesses:

- Assumes word independence
- Cannot capture contextual word usage or order
- Slight tendency toward false positives

Linear SVM**Strengths:**

- Best overall performance (highest F1 and AUC)
- Excellent class separability
- Robust against overfitting in sparse data

Weaknesses:

- Training becomes slower with very large feature sets
- No inherent sequence modeling capability

Bi-LSTM**Strengths:**

- Captures word order and long-range dependencies
- Competitive precision (on par with SVM)

Weaknesses:

- Requires significantly more training time (~10 minutes per epoch)
- Prone to overfitting without regularization
- Needs GPU resources for efficient training

6. Conclusion

In this project, we tackled the binary sentiment classification problem on the IMDB movie reviews dataset using three different modeling approaches: **Multinomial Naive Bayes**, **Linear SVM**, and a **Bidirectional LSTM (Bi-LSTM)**. Each model was implemented using appropriate text representations—TF-IDF vectors for Naive Bayes and SVM, and padded word-token sequences for the Bi-LSTM.

After extensive evaluation on a held-out test set of 10,000 reviews, **Linear SVM emerged as the most effective model**, achieving a **macro F1-score of 0.9036** and a **ROC-AUC of 0.9637**. These results reflect its strong generalization ability and superior decision boundary in high-dimensional TF-IDF space.

The **Multinomial Naive Bayes** classifier, while conceptually simple and computationally efficient, delivered robust performance ($F1 = 0.8774$) and served as a strong baseline. The **Bi-LSTM** model performed competitively as well ($F1 = 0.8803$), showing its strength in sequence modeling but also revealing its dependency on careful tuning and higher computational cost.

Overall, this study demonstrates that while deep learning models offer flexibility and contextual modeling capacity, traditional methods like SVM can outperform them when paired with well-engineered features. For future work, incorporating contextual word embeddings or performing hyperparameter optimization on the Bi-LSTM model could further improve results.

This project highlights the importance of model selection, feature representation, and evaluation methodology in building effective sentiment analysis systems.

7. Appendix

7.1 Team Contributions

<u>Team Member</u>	<u>Student ID</u>	<u>Contribution</u>
Onur Deniz Ögüncü	30744	Implemented statistical analysis of the dataset: word/character counts, vocabulary size, hapax ratio, and top-k words. Helped organize and verify summary table generation.
Defne Kızılkaya	31279	Developed TF-IDF preprocessing, trained and evaluated Multinomial Naive Bayes and Linear SVM models, contributed to classification report analysis, and co-wrote the final report.
Göktuğ Arda Gök	30975	Designed and implemented the Bi-LSTM model architecture in Keras, handled sequence padding, trained the deep learning model, and generated its predictions.
Uygar Hatipoğlu	30737	Handled dataset splitting, model evaluation metrics (accuracy, F1, ROC-AUC), produced classification reports and ROC curve plots for all models.
Muhammed Reşid Sağlam	31320	Managed file saving/loading processes for preprocessed data, configured tokenizer, ensured end-to-end reproducibility, and co-wrote and edited the final report.

7.2 Additional Tables/Graphs

```

=== Multinomial NB ===
      precision    recall  f1-score   support

   neg       0.88      0.87      0.88      5000
   pos       0.87      0.89      0.88      5000

 accuracy          0.88      10000
 macro avg       0.88      0.88      0.88      10000
weighted avg       0.88      0.88      0.88      10000

=== Linear SVM ===
      precision    recall  f1-score   support

   neg       0.91      0.89      0.90      5000
   pos       0.89      0.92      0.90      5000

 accuracy          0.90      10000
 macro avg       0.90      0.90      0.90      10000
weighted avg       0.90      0.90      0.90      10000

=== Bi-LSTM ===
      precision    recall  f1-score   support

   neg       0.87      0.89      0.88      5000
   pos       0.89      0.87      0.88      5000

 accuracy          0.88      10000
 macro avg       0.88      0.88      0.88      10000
weighted avg       0.88      0.88      0.88      10000

```

```

NB - Accuracy: 0.8763
NB - Precision: 0.8699
NB - Recall:    0.8850
NB - F1-score: 0.8774
NB - ROC-AUC:  0.9467

```

```

SVM - Accuracy: 0.9022
SVM - Precision: 0.8906
SVM - Recall:    0.9170
SVM - F1-score:  0.9036
SVM - ROC-AUC:   0.9637

```

```

Bi-LSTM - Accuracy: 0.8816
Bi-LSTM - Precision: 0.8903
Bi-LSTM - Recall:    0.8704
Bi-LSTM - F1-score:  0.8803
Bi-LSTM - ROC-AUC:   0.9508

```